

Transport-Calibrated Kernel Mean Embeddings for Robust Active Learning on Distributions

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Abstract—Active learning on distributions is a finite-sample decision problem in which each candidate query is a probability measure observed only through a bag of samples. Kernel mean embedding (KME) methods make this problem tractable by representing each bag as a Hilbert-space mean element and by using embedding uncertainty in the query rule. This paper gives KME-based probabilistic minimax active learning (PMAL) a transport-robust interpretation without replacing the implemented KME machinery by Wasserstein computation. When the RKHS feature map is Lipschitz, every Wasserstein perturbation induces a controlled KME perturbation; therefore a KME uncertainty set is a conservative Hilbert envelope of a transport ambiguity set. We derive finite-bag and Bayesian radii for this envelope, connect them to the reliability term in PMAL, and state an implementable transport-calibrated KME query rule balancing label uncertainty, distributional coverage, and embedding reliability. A 50-repeat benchmark suite covering Gaussian distributional tasks, hard-overlap Gaussian tasks, transformation-induced digit distributions, and a 20 Newsgroups distributional task shows that the calibrated rule improves over the finite-bag PMARLDS-lite baseline on the hard robust setting and remains competitive with strong margin sampling while providing an explicit robust-learning interpretation of the uncertainty term.

Index Terms—Active learning, learning on distributions, kernel mean embedding, Wasserstein distance, distributionally robust learning, probabilistic minimax learning, computational intelligence.

I. INTRODUCTION

In many pattern analysis and machine intelligence problems, the elementary object to be labeled is not a single vector. A handwritten digit can be represented by a distribution of transformations, a document by a distribution of word vectors, a patient by a collection of measurements, an activity by a trajectory segment, and a sensor unit by a bag of observations. Such problems are usually modeled as *learning on distributions*: the i th example is a probability measure P_i on a ground space $\mathcal{X}$, and the learner observes only a finite bag $B_i = \{x_{ij}\}_{j=1}^{n_i}$ sampled from that measure. In active learning on distributions, the learner must choose which distributional object should be labeled next. This is more difficult than vectorial active learning because two uncertainties coexist: the label is unknown and the distributional example itself is only estimated from finite samples.

Kernel mean embedding (KME) is a natural representation for this setting. A probability measure P is mapped to a Hilbert-space element

$$\mu_P = \int_{\mathcal{X}} k(x, \cdot) dP(x) \in \mathcal{H}, \quad (1)$$

where k is a positive definite kernel and $\mathcal{H}$ is its reproducing kernel Hilbert space (RKHS). The empirical embedding $\hat{\mu}_P = n^{-1} \sum_{j=1}^n k(x_j, \cdot)$ is computable from the observed bag, and the distance

$$\text{MMD}_k(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}} \quad (2)$$

allows kernel methods to compare probability measures. Bayesian KME goes further by placing a posterior distribution on the unknown embedding and therefore quantifying uncertainty caused by finite bags. This uncertainty has already proved useful for robust active learning on distributions.

At the same time, recent machine learning has increasingly used optimal transport, Sinkhorn divergences, and Wasserstein distributionally robust optimization (DRO) to reason about geometry-aware uncertainty. This creates an important research opportunity. Instead of asking whether KME should be replaced by Wasserstein distance, we ask a more precise question:

Can Bayesian KME-based PMAL be understood as a tractable surrogate for an active learning problem whose uncertainty is defined on transport neighborhoods of probability measures?

This question is different from a simple change of metric. A direct substitution of $\|\mu_P - \mu_Q\|_{\mathcal{H}}$ by $W(P, Q)$ would make the theory and implementation inconsistent if the algorithm remains KME-based. The central idea of this paper is therefore *calibration*, not replacement. KME remains the computational representation, while Wasserstein geometry gives a modern interpretation of the finite-bag uncertainty that KME is controlling.

A. Why a New View Is Needed

In distributional active learning, not all candidates are equally reliable. A bag with five samples and a bag with one hundred samples may have similar empirical embeddings, but their uncertainty should not be treated equally. Bayesian KME captures this phenomenon by assigning an embedding posterior covariance or a confidence radius to each distributional example. From the perspective of contemporary robust learning, this radius is not merely a statistical artifact; it can be interpreted as a surrogate ambiguity radius around an unknown probability measure.

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Transport geometry is especially relevant because it compares distributions using the geometry of the underlying sample space. For example, small translations of images, slight shifts in word-vector clouds, and perturbations in sensor measurements are often better represented by transport displacement than by pointwise density divergence. Nevertheless, exact Wasserstein computation inside each active-learning iteration is expensive and would require a different algorithmic design. KME offers a scalable Hilbert-space surrogate. The missing piece is a theoretical bridge between the two views.

B. Core Insight

Let $W_1(P, Q)$ denote the 1-Wasserstein distance and let $\text{MMD}_k(P, Q)$ denote the KME distance. If the feature map $\phi(x) = k(x, \cdot)$ is Lipschitz, then

$$\text{MMD}_k(P, Q) \leq L_\phi W_1(P, Q). \quad (3)$$

Thus, every Wasserstein ball around an empirical distribution is mapped into a KME ball around its empirical embedding. Consequently, a KME uncertainty set can be used as a conservative Hilbert envelope for a transport ambiguity set. Under additional assumptions that make the kernel norm control Wasserstein distance on the considered model class, a reverse inequality also holds locally. This yields a two-sided calibration between transport and KME geometries.

This result is conceptually important. It shows that KME uncertainty and Wasserstein uncertainty are not universally identical, but they can be linked by explicit assumptions. Therefore, the Bayesian KME active-learning objective can be interpreted as a tractable upper envelope of a transport-robust PMAL objective. This is the new scientific angle developed in this paper.

C. Contributions

The paper makes the following contributions.

- It formulates distributional active learning as a post-query decision problem under finite-bag uncertainty on the space of probability measures.
- It introduces a transport-calibrated interpretation of Bayesian KME-PMAL, in which KME uncertainty is a conservative Hilbert envelope of Wasserstein ambiguity.
- It proves that the KME-PMAL robust score upper-bounds a Wasserstein robust post-query risk under a Lipschitz feature-map assumption.
- It derives finite-bag and Bayesian credible radii that connect bag size, posterior covariance, and active query reliability.
- It gives a two-sided calibration result under transport-controlling kernels, clarifying when KME and transport geometries are comparable and when they are not.
- It gives an explicit query algorithm and a reproducible benchmark suite with Gaussian, hard-overlap Gaussian, digit-distribution, and document-distribution tasks.

II. BACKGROUND AND RELATED WORK

A. Active Learning and Probabilistic Minimax Selection

Active learning reduces labeling cost by selecting unlabeled candidates whose labels are expected to improve the model

most. Common strategies include uncertainty sampling, margin sampling, query-by-committee, expected model change, expected error reduction, and representative sampling. These criteria often emphasize either informativeness or representativeness, but balancing the two is difficult.

Probabilistic minimax active learning (PMAL) addresses this difficulty by considering the risk after the queried label is obtained, before that label is known. The learner evaluates a candidate through a worst-case or minimax view of possible labels, together with an optimistic or probabilistic completion of the unknown labels of the remaining unlabeled data. The result is a query principle that is naturally robust to label uncertainty and can be implemented in Bayesian settings.

The present paper does not redevelop functional-gradient PMAL for vectorial data. Instead, it focuses on distributional examples and the uncertainty induced by finite bags. The central issue is not how to optimize a functional classifier, but how to interpret KME uncertainty as a geometry-aware robust ambiguity set.

B. Learning on Distributions and KME

Learning on distributions assumes that each object is a probability measure P_i and that the learner observes a finite sample from it. KME turns this problem into learning in an RKHS by mapping P_i to μ_{P_i} . If the kernel is characteristic, the embedding is injective on probability measures, so learning on embeddings is expressive. The empirical embedding has convergence guarantees, but its estimation error depends on bag size. Bayesian KME uses a Gaussian process prior to model uncertainty of the mean element and yields posterior covariance information.

In active learning, this posterior uncertainty matters. A candidate with high uncertainty may be informative, but it may also be unreliable. Bayesian KME-PMAL uses this uncertainty inside a probabilistic minimax query score. The present paper supplies a new interpretation: the posterior embedding uncertainty can be treated as a data-dependent robust uncertainty set that approximates a transport ambiguity set.

C. Wasserstein DRO and Transport Geometry

Wasserstein distance compares probability measures by the cost of transporting mass across the ground space. For W_1 , the Kantorovich-Rubinstein duality states

$$W_1(P, Q) = \sup_{\text{Lip}(f) \leq 1} \left\{ \int f dP - \int f dQ \right\}. \quad (4)$$

Wasserstein DRO uses ambiguity sets of the form $\{Q : W(Q, \hat{P}) \leq \rho\}$ to optimize decisions that are robust to distributional perturbations. This is attractive in learning on distributions because empirical bags are finite and uncertain.

However, exact transport-based active learning is expensive. Each candidate query would require solving a robust post-query problem over probability measures. KME makes the problem tractable by replacing probability measures with Hilbert-space embeddings. The central question is whether this replacement is only computational or also theoretically justified. The calibration results in this paper answer this question.

D. Sinkhorn Divergence as a Bridge

Entropy-regularized optimal transport and Sinkhorn divergence provide a smooth and scalable bridge between OT and MMD. When the regularization is small, Sinkhorn divergence is closer to OT; with strong regularization, it behaves more like a kernel discrepancy. This makes Sinkhorn geometry a natural language for explaining why KME-based methods can serve as tractable endpoints of a broader transport-geometric family. In this paper, Sinkhorn is used as a theoretical bridge, not as a new experimental method.

III. PROBLEM FORMULATION

Let $(\mathcal{X}, d)$ be a Polish metric space and let $\mathcal{P}(\mathcal{X})$ denote probability measures on $\mathcal{X}$ with finite first moment. A distributional classification dataset is

$$\mathcal{D} = \{(B_i, y_i)\}_{i=1}^N, \quad (5)$$

where $y_i \in \mathcal{Y} = \{1, \dots, C\}$ and

$$B_i = \{x_{ij}\}_{j=1}^{n_i}, \quad x_{ij} \sim P_i. \quad (6)$$

The measure P_i is unobserved. Its empirical version is

$$\hat{P}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \delta_{x_{ij}}. \quad (7)$$

At an active-learning stage, the index set is divided into labeled and unlabeled subsets L and U . The learner chooses $q \in U$ and receives y_q .

Let h_L denote a classifier trained with labeled distributional examples indexed by L . The ideal risk is

$$R(h_L) = \mathbb{E}_{(P,Y)} \ell(h_L(P), Y), \quad (8)$$

where ℓ is a bounded loss. Because P is not exactly observed, the empirical classifier uses representations of $\hat{P}_i$. In KME learning,

$$\hat{\mu}_i = \mu_{\hat{P}_i} = \frac{1}{n_i} \sum_{j=1}^{n_i} k(x_{ij}, \cdot). \quad (9)$$

The post-query risk of candidate q is label-dependent. Since y_q is unknown before querying, PMAL evaluates an upper envelope of the risk after the query. A generic PMAL score has the form

$$\mathcal{Q}(q) = \max_{y_q \in \mathcal{Y}} \mathcal{B}(h_{L \cup \{q\}}, \mathcal{D}_L, (B_q, y_q), \mathcal{D}_U), \quad (10)$$

where $\mathcal{B}$ is a computable post-query risk bound. Distributional uncertainty modifies this score by replacing each empirical distribution with an uncertainty set.

Definition 1 (Distributional active-learning ambiguity). *For each bag B_i , an ambiguity map assigns a set $\mathcal{A}_i$ of plausible probability measures such that $P_i \in \mathcal{A}_i$ with high probability. The robust post-query score is*

$$\mathcal{Q}_{\mathcal{A}}(q) = \max_{y_q \in \mathcal{Y}} \sup_{\{Q_i \in \mathcal{A}_i\}} \mathcal{B}(h_{L \cup \{q\}}, \{(Q_i, y_i) : i \in L \cup \{q\}\}, \{Q_j : j \in U \setminus \{q\}\}). \quad (11)$$

The main question is how to choose $\mathcal{A}_i$. The transport view suggests Wasserstein balls. The KME implementation suggests Hilbert balls or Bayesian credible ellipsoids. This paper calibrates the two.

A. The Master Minimax Problem Behind the Article

The actual problem solved by a distributional PMAL method is not a single heuristic score. It is a nested post-query decision problem. Before querying q , the learner does not know its label; it also does not know the labels of the remaining unlabeled bags; and, in learning on distributions, it does not observe the true measures P_i but only finite bags. A compact way to express the ideal transport-robust problem is

$$q_t^* \in \arg \min_{q \in U_t} \max_{y \in \mathcal{Y}} \min_{h \in \mathcal{F}} \min_{\tilde{y} \in \mathcal{Y}^{|U_t|-1}} \sup_{\mathbf{Q} \in \mathfrak{B}_W(\rho)} \mathcal{J}_t(h; \mathbf{Q}, \mathbf{y}_L, (q, y), \tilde{\mathbf{y}}), \quad (12)$$

where

$$\mathfrak{B}_W(\rho) = \prod_{i=1}^N \{Q_i : W_1(Q_i, \hat{P}_i) \leq \rho_i\}. \quad (13)$$

The five operators in (12) have distinct roles: query selection is an outer minimization; the queried label is adversarial because it is unknown before the oracle answers; the classifier is trained by risk minimization; the remaining unlabeled labels are completed optimistically as in PMAL; and the finite-bag distributional error is handled by a robust transport supremum. This is the exact place where the thesis-style “min–max–min–min–max” structure enters the new article.

The direct problem in (12) is not the algorithm we run. It is the target robust decision problem. The contribution of this paper is to prove and implement the following reduction chain:

$$\begin{aligned} & \text{transport PMAL master problem} \\ & \Downarrow \text{Lipschitz KME calibration} \\ & \text{Hilbert-ball PMAL upper envelope} \\ & \Downarrow \text{finite-bag/Bayesian KME radius} \\ & \text{computable TC-BKME-PMAL query rule.} \end{aligned} \quad (14)$$

Thus the paper is not merely a reinterpretation. It states the original minimax object, proves why a KME relaxation is a conservative surrogate, and gives code that evaluates the resulting query rule on the same benchmark family as the thesis experiments.

Proposition 1 (Hilbert relaxation of the master problem). *Assume the feature-map Lipschitz condition in Assumption 1 and set $\tau_i = L_\phi \rho_i$. Replacing $\mathfrak{B}_W(\rho)$ in (12) by*

$$\mathfrak{B}_H(\tau) = \prod_{i=1}^N \{Q_i : \|\mu_{Q_i} - \mu_{\hat{P}_i}\|_{\mathcal{H}} \leq \tau_i\} \quad (15)$$

yields a conservative upper envelope of the transport-robust PMAL objective.

Proof. The set inclusion in Corollary 1 holds for each bag. Taking the product over bags gives $\mathfrak{B}_W(\rho) \subseteq \mathfrak{B}_H(\tau)$. The supremum over the larger product set is therefore no smaller, and all outer minimization, label maximization, and classifier minimization operators preserve the conservative envelope interpretation. $\square$

IV. TRANSPORT AND HILBERT AMBIGUITY SETS

A. Transport Ambiguity

For a radius $\rho_i > 0$, define the Wasserstein ambiguity set

$$\mathbb{B}_W(\hat{P}_i, \rho_i) = \{Q \in \mathcal{P}(\mathcal{X}) : W_1(Q, \hat{P}_i) \leq \rho_i\}. \quad (16)$$

This set is geometry-aware: it contains distributions that can be reached by moving the empirical mass by a small average distance.

B. Hilbert Ambiguity

For a radius $\tau_i > 0$, define the KME ambiguity set

$$\mathbb{B}_H(\hat{P}_i, \tau_i) = \{Q \in \mathcal{P}(\mathcal{X}) : \|\mu_Q - \mu_{\hat{P}_i}\|_{\mathcal{H}} \leq \tau_i\}. \quad (17)$$

This set is computationally compatible with kernel learning. In a Bayesian KME implementation, $\mathbb{B}_H$ can be replaced by a posterior credible ellipsoid

$$\mathcal{E}_i(\delta) = \{z : (z - m_i)^T \Sigma_i^\dagger (z - m_i) \leq c_\delta\}, \quad (18)$$

where m_i and Σ_i are posterior mean and covariance of the embedding representation.

C. Why Direct Replacement Is Wrong

It is tempting to replace every KME distance by a Wasserstein distance. This is not generally valid for three reasons. First, KME and Wasserstein measure different aspects of distributions. Second, the KME-based posterior covariance is defined in a Hilbert space, not in the nonlinear Wasserstein space. Third, existing experiments based on KME would no longer directly validate a purely Wasserstein algorithm. Therefore, the correct strategy is not replacement but calibration: use transport to define the desired robust geometry and use KME to compute a tractable conservative envelope.

V. TRANSPORT-TO-KME CALIBRATION

Assumption 1 (Lipschitz RKHS feature map). *The canonical feature map $\phi(x) = k(x, \cdot)$ is L_ϕ -Lipschitz:*

$$\|\phi(x) - \phi(x')\|_{\mathcal{H}} \leq L_\phi d(x, x'), \quad \forall x, x' \in \mathcal{X}. \quad (19)$$

Lemma 1 (Wasserstein perturbations induce KME perturbations). *Under Assumption 1, for all $P, Q \in \mathcal{P}(\mathcal{X})$,*

$$\text{MMD}_k(P, Q) \leq L_\phi W_1(P, Q). \quad (20)$$

Proof. By the variational representation of MMD,

$$\text{MMD}_k(P, Q) = \sup_{\|f\|_{\mathcal{H}} \leq 1} \left| \int f dP - \int f dQ \right|. \quad (21)$$

For any $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} \leq 1$,

$$|f(x) - f(x')| = |\langle f, \phi(x) - \phi(x') \rangle| \leq L_\phi d(x, x'), \quad (22)$$

so f/L_ϕ is 1-Lipschitz. The Kantorovich-Rubinstein duality then gives the result. $\square$

Corollary 1 (Set inclusion). *Under Assumption 1,*

$$\mathbb{B}_W(\hat{P}_i, \rho_i) \subseteq \mathbb{B}_H(\hat{P}_i, L_\phi \rho_i). \quad (23)$$

Proof. If $Q \in \mathbb{B}_W(\hat{P}_i, \rho_i)$, then $\text{MMD}_k(Q, \hat{P}_i) \leq L_\phi \rho_i$ by Lemma 1. Thus $Q \in \mathbb{B}_H(\hat{P}_i, L_\phi \rho_i)$. $\square$

A. Reverse Calibration on Restricted Model Classes

The reverse inequality is not true for arbitrary kernels and arbitrary measures. The correct statement must be restricted.

Assumption 2 (Transport-controlling kernel on a model class). *There exists a model class $\mathcal{P}_0 \subset \mathcal{P}(\mathcal{X})$ and constants $C_k > 0$ and $\alpha \in (0, 1]$ such that*

$$W_1(P, Q) \leq C_k \text{MMD}_k(P, Q)^\alpha, \quad \forall P, Q \in \mathcal{P}_0. \quad (24)$$

Proposition 2 (Two-sided calibration). *If Assumptions 1 and 2 hold, then for all $P, Q \in \mathcal{P}_0$,*

$$\frac{1}{L_\phi} \text{MMD}_k(P, Q) \leq W_1(P, Q) \leq C_k \text{MMD}_k(P, Q)^\alpha. \quad (25)$$

Thus the transport and KME geometries are locally comparable on $\mathcal{P}_0$.

Remark 1. The word “calibration” is used intentionally. KME and Wasserstein distances are not the same. The two-sided result requires a restricted distribution class and a kernel with enough resolution. The conservative one-sided result in Corollary 1 is the main tool for robust active learning.

VI. FINITE-BAG AND BAYESIAN RADII

A. Empirical KME Radius

Assume $k(x, x) \leq \kappa^2$. The empirical embedding is an average of Hilbert-valued random variables. A standard concentration result implies that, with probability at least $1 - \delta$,

$$\|\mu_{P_i} - \mu_{\hat{P}_i}\|_{\mathcal{H}} \leq r_i(\delta) = c\kappa \sqrt{\frac{\log(1/\delta)}{n_i}}, \quad (26)$$

where c is a universal constant depending on the chosen concentration inequality. The exact constant is less important than the dependence $n_i^{-1/2}$.

This result provides the simplest KME ambiguity radius:

$$\tau_i = r_i(\delta_i). \quad (27)$$

By Corollary 1, the corresponding implicit Wasserstein radius is

$$\rho_i = \tau_i / L_\phi. \quad (28)$$

Thus finite-bag uncertainty is converted into a transport-calibrated robust set.

B. Bayesian KME Radius

Bayesian KME gives a sharper and anisotropic uncertainty description. Suppose that after observing B_i , the embedding representation is approximately Gaussian:

$$z_i = \mu_{P_i} \mid B_i \approx \mathcal{N}(m_i, \Sigma_i). \quad (29)$$

Then a credible ellipsoid is

$$\mathcal{E}_i(\delta) = \{z : (z - m_i)^T \Sigma_i^\dagger (z - m_i) \leq \chi_{r, 1-\delta}^2\}, \quad (30)$$

where r is the effective rank. A conservative isotropic radius is

$$\tau_i^{\text{Bayes}}(\delta) = \sqrt{\lambda_{\max}(\Sigma_i) \chi_{r, 1-\delta}^2}. \quad (31)$$

The key difference between empirical and Bayesian radii is that the Bayesian radius depends not only on n_i but also on the geometry of observed samples and the posterior covariance direction.

Proposition 3 (Reliability-weighted embedding uncertainty). *For a linear classifier $f_w(z) = w^T z$ and an L_ℓ -Lipschitz scalar loss, the worst-case loss over the Bayesian credible ball satisfies*

$$\sup_{\|z - m_i\| \leq \tau_i} \ell(f_w(z), y_i) \leq \ell(f_w(m_i), y_i) + L_\ell \|w\| \tau_i. \quad (32)$$

For an ellipsoid $\mathcal{E}_i(\delta)$, the sharper bound is

$$\sup_{z \in \mathcal{E}_i(\delta)} \ell(f_w(z), y_i) \leq \ell(f_w(m_i), y_i) + L_\ell \sqrt{\chi_{r,1-\delta}^2} \sqrt{w^T \Sigma_i w}. \quad (33)$$

Proof. The ball case follows from Lipschitz continuity and Cauchy-Schwarz. For the ellipsoid, maximize $w^T(z - m_i)$ subject to $(z - m_i)^T \Sigma_i^\dagger (z - m_i) \leq \chi_{r,1-\delta}^2$, which yields the support function of the ellipsoid. $\square$

VII. TRANSPORT-CALIBRATED PMAL OBJECTIVE

A. Transport-Robust Post-Query Risk

For a candidate query $q \in U$, define the ideal transport-robust PMAL score

$$\mathcal{Q}_W(q) = \max_{y_q \in \mathcal{Y}} \sup_{\{Q_i \in \mathbb{B}_W(\hat{P}_i, \rho_i)\}} \mathcal{B}(h_{L \cup \{q\}}, \{(Q_i, y_i)\}_{i \in L \cup \{q\}}, \{Q_j\}_{j \in U \setminus \{q\}}). \quad (34)$$

This is the robust objective that we would like to minimize if transport computation were cheap and if all post-query label cases could be handled exactly.

B. KME Surrogate Post-Query Risk

The KME surrogate replaces $\mathbb{B}_W$ with $\mathbb{B}_H$:

$$\mathcal{Q}_H(q) = \max_{y_q \in \mathcal{Y}} \sup_{\{Q_i \in \mathbb{B}_H(\hat{P}_i, \tau_i)\}} \mathcal{B}(h_{L \cup \{q\}}, \{(Q_i, y_i)\}_{i \in L \cup \{q\}}, \{Q_j\}_{j \in U \setminus \{q\}}). \quad (35)$$

The implemented KME-PMAL method computes a tractable bound or approximation to $\mathcal{Q}_H(q)$ using kernel matrices and Bayesian embedding covariance.

Theorem 1 (Conservative transport-calibrated PMAL). *Assume Assumption 1. Set $\tau_i = L_\phi \rho_i$ for every bag. Then, for every candidate query $q \in U$,*

$$\mathcal{Q}_W(q) \leq \mathcal{Q}_H(q). \quad (36)$$

Therefore, minimizing the KME-PMAL surrogate minimizes an upper envelope of the transport-robust post-query score.

Proof. By Corollary 1, every feasible distribution in the Wasserstein ambiguity set is also feasible in the KME ambiguity set. The supremum over the KME set is therefore no smaller. The maximization over y_q preserves the inequality. $\square$

C. Excess Robust Score Bound

Let $q_W^* = \arg \min_q \mathcal{Q}_W(q)$ and $q_H^* = \arg \min_q \mathcal{Q}_H(q)$. Since $\mathcal{Q}_H$ is an upper bound, q_H^* is not necessarily identical to q_W^* . The following result quantifies the effect of calibration looseness.

Definition 2 (Calibration gap). *For a candidate q , define*

$$\Delta(q) = \mathcal{Q}_H(q) - \mathcal{Q}_W(q) \geq 0. \quad (37)$$

Let $\Delta_{\max} = \max_{q \in U} \Delta(q)$.

Theorem 2 (Regret induced by surrogate calibration). *The transport-robust score of the KME-selected query satisfies*

$$\mathcal{Q}_W(q_H^*) - \mathcal{Q}_W(q_W^*) \leq \Delta(q_W^*). \quad (38)$$

In particular,

$$\mathcal{Q}_W(q_H^*) - \mathcal{Q}_W(q_W^*) \leq \Delta_{\max}. \quad (39)$$

Proof. Because q_H^* minimizes $\mathcal{Q}_H$,

$$\mathcal{Q}_H(q_H^*) \leq \mathcal{Q}_H(q_W^*) = \mathcal{Q}_W(q_W^*) + \Delta(q_W^*). \quad (40)$$

Also, $\mathcal{Q}_W(q_H^*) \leq \mathcal{Q}_H(q_H^*)$. Combining the two inequalities gives the result. $\square$

This theorem explains the desired scientific position. The KME method is defensible as long as the calibration gap is controlled. The role of kernel choice, Bayesian covariance, and bag-size radius is precisely to make this gap small.

D. Active Selection Rule

The proposed query rule is

$$q^* \in \arg \min_{q \in U} \hat{\mathcal{Q}}_H(q), \quad (41)$$

where $\hat{\mathcal{Q}}_H$ is the computable PMAL bound based on Bayesian kernel embeddings. The word “transport-calibrated” means that the radii and uncertainty terms are chosen so that the Hilbert ambiguity set covers or approximates a transport ambiguity set.

E. Solved Finite-Dimensional Surrogate

To avoid the impression that the paper stops at an abstract inclusion theorem, we now state the finite-dimensional surrogate that is actually solved in the code. Let $z_i = \hat{\mu}_i$ be an empirical or random-feature KME representation and let $f_w(P_i) = w^T z_i$ be a linear classifier in the embedding space. For a Lipschitz convex loss, the Hilbert-ball robustification gives the tractable upper bound

$$\begin{aligned} \hat{\mathcal{J}}_H(w) = & \sum_{i \in L_t} \ell(y_i, w^T z_i) + \lambda \|w\|_2^2 \\ & + \gamma \sum_{i \in L_t} \tau_i \|w\|_2. \end{aligned} \quad (42)$$

For a candidate query q , the first-order PMAL reduction evaluates how the candidate changes the robust upper envelope after

both possible labels. This yields the computable acquisition rule

$$S_i^{\text{TC}} = \underbrace{(\epsilon_m + |f_L(z_i)|)^{-1}}_{\text{label uncertainty}} \underbrace{(1 + \alpha D_i)}_{\text{distributional coverage}} \times \underbrace{(1 + \eta \tau_i)^{-1}}_{\text{transport-calibrated reliability}}, \quad (43)$$

where

$$D_i = \frac{1}{|U_t| - 1} \sum_{j \in U_t, j \neq i} \exp\left(-\frac{\|z_i - z_j\|_2^2}{2s_U^2}\right). \quad (44)$$

The active learner queries $\arg \max_{i \in U_t} S_i^{\text{TC}}$. This rule is not presented as the exact optimizer of (12); it is the explicit finite-dimensional greedy solution of the calibrated upper envelope. This distinction makes the method implementable while keeping the minimax origin visible.

F. First-Order Robust Envelope Behind the Practical Score

The practical acquisition score in (43) should not appear suddenly after the minimax formulation. It is obtained by a first-order robust envelope of the Hilbert problem. Let Φ_i denote the embedded representation of the i th bag and let $\mathcal{B}_t(\Phi_1, \dots, \Phi_N; q, y)$ be the post-query PMAL bound for candidate q and hypothetical label y . Around the empirical embeddings $\hat{\Phi}_i = \hat{\mu}_i$, the Hilbert robust envelope is

$$\bar{\mathcal{B}}_t(q, y) = \sup_{\|\Phi_i - \hat{\Phi}_i\|_{\mathcal{H}} \leq \tau_i} \mathcal{B}_t(\Phi_1, \dots, \Phi_N; q, y). \quad (45)$$

If $\mathcal{B}_t$ is locally differentiable with respect to the embeddings, then Cauchy's inequality gives

$$\bar{\mathcal{B}}_t(q, y) \leq \mathcal{B}_t(\hat{\Phi}_1, \dots, \hat{\Phi}_N; q, y) + \sum_{i=1}^N \tau_i \|\nabla_{\Phi_i} \mathcal{B}_t\|_{\mathcal{H}} + o(\tau). \quad (46)$$

The first term is the nominal PMAL score. The gradient norm measures how sensitive the score is to perturbing each bag. The radius τ_i measures how unreliable the empirical bag representation is. Thus, the robust correction is not an arbitrary penalty; it is the product of sensitivity and finite-bag uncertainty.

Theorem 3 (Local finite-dimensional reduction). *Assume that the post-query bound is locally differentiable in the KME variables and that the classifier is linear in a finite random-feature approximation z_i of μ_i . Then the Hilbert robust PMAL envelope admits the local upper bound*

$$\max_y \bar{\mathcal{B}}_t(q, y) \leq \max_y \mathcal{B}_t(\mathbf{z}; q, y) + C_t \tau_q + \sum_{j \in U_t \setminus \{q\}} C_{tj} \tau_j + o(\tau), \quad (47)$$

where the constants C_t and C_{tj} are sensitivity coefficients determined by the current classifier, loss slope, and kernel-feature norm.

Proof. Apply (46) to each hypothetical label y and then take the maximum over y . In the finite random-feature representation, the Frechet derivative in the RKHS is represented by the

Algorithm 1 Transport-Calibrated Bayesian KME-PMAL

Require: Labeled bags $\mathcal{D}_L$, unlabeled bags $\mathcal{D}_U$, kernel k , confidence level δ , feature-map Lipschitz constant or estimate L_ϕ .

- 1: For every bag B_i , compute empirical KME $\hat{\mu}_i$.
- 2: Estimate Bayesian KME posterior mean m_i and covariance Σ_i , or use empirical radius $r_i(\delta)$.
- 3: Set Hilbert radius τ_i using (26) or (31).
- 4: Interpret the implicit transport radius as $\rho_i = \tau_i / L_\phi$ for conservative calibration.
- 5: **for** each candidate query $q \in U$ **do**
- 6: Evaluate the PMAL post-query upper bound using m_i and the uncertainty term induced by τ_i or Σ_i .
- 7: Store the robust score $\hat{\mathcal{Q}}_H(q)$.
- 8: **end for**
- 9: Query $q^* = \arg \min_{q \in U} \hat{\mathcal{Q}}_H(q)$.
- 10: Obtain y_{q^*} and update $\mathcal{D}_L \leftarrow \mathcal{D}_L \cup \{(B_{q^*}, y_{q^*})\}$.

Euclidean gradient with respect to z_i . Boundedness of the loss slope and the classifier norm gives finite constants C_t and C_{tj} . The $o(\tau)$ term follows from local differentiability. $\square$

Theorem 3 explains the three factors in the executable rule. The nominal label-dependent PMAL term contributes the uncertainty factor; the unlabeled-pool sensitivity contributes the representativeness or coverage factor; and the radius term contributes the reliability correction. The code uses the multiplicative form in (43) because it is scale-stable across datasets, while the theorem justifies the same components from the additive robust envelope.

VIII. CONNECTION TO THE PMARLDB QUADRATIC FORM

Bayesian KME active learning often yields a quadratic label-dependent objective of the form

$$\hat{P}(\mathbf{y} \mid \hat{\mu}) \propto \exp\left(-\frac{1}{2} \mathbf{y}^T S \mathbf{y}\right), \quad (48)$$

where S is computed from kernel matrices, posterior covariance terms, and regularization. The PMAL query is obtained by evaluating how a candidate label changes this quadratic post-query bound.

The transport-calibrated view gives a new meaning to S . It is not only a kernel similarity matrix. It is an operator encoding how uncertainty in finite-bag embeddings changes label risk under a robust envelope. If S is decomposed as

$$S = S_{\text{sim}} + S_{\text{unc}}, \quad (49)$$

then S_{sim} captures similarity among embedded distributions, while S_{unc} captures reliability penalties caused by posterior covariance. Transport calibration interprets S_{unc} as the Hilbert representation of an ambiguity radius in measure space.

Proposition 4 (Quadratic robust penalty). *Assume a Gaussian posterior embedding model and a linear Gaussian label model in the embedding space. Marginalizing the embedding uncertainty yields an effective label likelihood with variance*

$$\sigma_i^2(w) = \sigma^2 + w^T \Sigma_i w. \quad (50)$$

The negative log-likelihood is upper-bounded by a nominal quadratic loss plus a reliability penalty proportional to $w^T \Sigma_i w$. Under the transport calibration $\tau_i^2 \asymp \lambda_{\max}(\Sigma_i)$, this penalty is an empirical Hilbert-space surrogate of finite-bag transport ambiguity.

Proof. The Gaussian marginalization gives

$$y_i \mid m_i, w \sim \mathcal{N}(w^T m_i, \sigma^2 + w^T \Sigma_i w). \quad (51)$$

The negative log-likelihood contains

$$\frac{(y_i - w^T m_i)^2}{\sigma^2 + w^T \Sigma_i w} + \log(\sigma^2 + w^T \Sigma_i w). \quad (52)$$

Upper bounding and regularizing this expression yields a quadratic nominal part and a covariance-dependent penalty. Since $\lambda_{\max}(\Sigma_i)$ controls the credible radius, the penalty is controlled by the same quantity that defines the Hilbert ambiguity set. $\square$

IX. SINKHORN-CALIBRATED INTERPRETATION

The calibration above uses W_1 and MMD. Sinkhorn divergence provides an intermediate family. For a ground cost c and regularization $\varepsilon > 0$, entropy-regularized OT is

$$\text{OT}_\varepsilon(P, Q) = \inf_{\pi \in \Pi(P, Q)} \int c(x, z) d\pi(x, z) + \varepsilon \text{KL}(\pi \parallel P \otimes Q). \quad (53)$$

The debiased Sinkhorn divergence is

$$S_\varepsilon(P, Q) = \text{OT}_\varepsilon(P, Q) - \frac{1}{2} \text{OT}_\varepsilon(P, P) - \frac{1}{2} \text{OT}_\varepsilon(Q, Q). \quad (54)$$

For small ε , S_ε behaves closer to transport. For large ε , it approaches a smooth kernel-like discrepancy. Therefore, one can define a family of ambiguity sets

$$\mathbb{B}_{S_\varepsilon}(\hat{P}_i, \rho_i) = \{Q : S_\varepsilon(Q, \hat{P}_i) \leq \rho_i\}. \quad (55)$$

This family provides a continuous conceptual path:

$$\begin{aligned} \text{Wasserstein ambiguity} &\longleftrightarrow \text{Sinkhorn ambiguity} \\ &\longleftrightarrow \text{KME/MMD ambiguity}. \end{aligned} \quad (56)$$

Proposition 5 (Regularized geometric path). *Let $\mathcal{X}$ be compact and c continuous. The family S_ε gives differentiable geometric divergences that interpolate between OT-like and MMD-like behavior as the regularization strength varies. Consequently, KME-PMAL can be interpreted as the computational endpoint of a regularized transport-PMAL family.*

Remark 2. *This proposition is not an experimental claim. It is a conceptual bridge that justifies why a modern transport-geometric paper can keep the original KME computations. A full Sinkhorn-PMAL algorithm would require new experiments; this manuscript intentionally avoids that claim.*

X. FROM TRANSPORT CALIBRATION TO AN IMPLEMENTABLE QUERY ALGORITHM

The preceding sections justify a transport-calibrated interpretation of Bayesian KME uncertainty. For a publishable journal manuscript, however, this interpretation must lead to an algorithm that can be implemented and evaluated on the

same benchmarks used for distributional active learning. This section makes the computational object explicit.

The full Bayesian KME-PMAL objective can be expensive because it propagates posterior embedding covariance through a post-query minimax criterion. For experimental implementation, we use the following surrogate decomposition. For an unlabeled distributional candidate i , define

$$u_i = \frac{1}{\epsilon_m + |f_L(\hat{\mu}_i)|}, \quad (57)$$

$$d_i = \frac{1}{|U| - 1} \sum_{j \in U, j \neq i} \exp\left(-\frac{\|\hat{\mu}_i - \hat{\mu}_j\|_{\mathcal{H}}^2}{2s_U^2}\right), \quad (58)$$

$$a_i = \frac{1}{1 + \eta\tau_i}, \quad (59)$$

where u_i is a label-uncertainty factor, d_i is a distributional representativeness factor, a_i is a transport-calibrated reliability factor, f_L is the classifier trained on currently labeled distributional examples, s_U is a median bandwidth in embedding space, and τ_i is the Hilbert ambiguity radius induced by finite-bag or Bayesian embedding uncertainty. The query score is

$$S_i^{\text{TC}} = u_i (1 + \alpha d_i) a_i. \quad (60)$$

The factor a_i is the new transport-calibrated part of the score: using Lemma 1, a Hilbert radius τ_i corresponds to a conservative transport radius $\rho_i = \tau_i/L_\phi$. Thus, candidates whose empirical distributions are poorly estimated are not over-trusted even if their empirical embedding lies close to the current boundary.

For a bounded kernel, a finite-bag radius may be set as

$$\tau_i(\delta) = \kappa \sqrt{\frac{\log(2N/\delta)}{n_i}}, \quad (61)$$

where κ absorbs kernel and confidence constants. In the full Bayesian KME version, this scalar radius can be replaced by a covariance-shaped credible radius

$$\tau_i^2(\delta) = \chi_{r,1-\delta}^2 \lambda_{\max}(\Sigma_i), \quad (62)$$

or by the full ellipsoidal score obtained from Σ_i . The scalar version is used in the reproducible benchmark code; the covariance version gives the corresponding Bayesian extension.

Remark 3 (Relation to the original PMARLDB computation). *Algorithm 2 is not a new Sinkhorn active-learning algorithm. It is an implementation-oriented transport-calibrated KME rule. The scalar radius in (61) can be replaced by a Bayesian embedding covariance when a full PMARLDB implementation is available; the benchmark implementation uses the finite-bag radius to keep the protocol reproducible.*

XI. EXPERIMENTAL PROTOCOL AND EXECUTABLE BENCHMARK SUITE

A high-level theoretical reinterpretation is not sufficient for a strong journal submission. The proposed transport-calibrated view must produce a query rule that can be inserted into the same distributional benchmark pipeline used in the thesis experiments. This section therefore turns the theory into an executable protocol. The main principle is simple: keep the

Algorithm 2 Transport-Calibrated Bayesian KME-PMAL (TC-BKME-PMAL)

Require: Bags $\{B_i\}_{i=1}^N$, initial labels L , unlabeled set U , kernel k , budget B , parameters α, η, δ .

- 1: Compute empirical KME representations $\hat{\mu}_i = n_i^{-1} \sum_{x \in B_i} k(x, \cdot)$.
- 2: Estimate ambiguity radii τ_i by (61) or posterior covariance radii by (62).
- 3: **for** $t = 1, \dots, B$ **do**
- 4: Train a distributional classifier f_L on $\{(\hat{\mu}_i, y_i) : i \in L\}$ using an empirical expectation kernel or its random-feature approximation.
- 5: Compute the embedding-space median bandwidth s_U over $\{\hat{\mu}_i : i \in U\}$.
- 6: **for** each $i \in U$ **do**
- 7: Compute uncertainty $u_i = (\epsilon_m + |f_L(\hat{\mu}_i)|)^{-1}$.
- 8: Compute representativeness $d_i = \frac{|U|^{-1} \sum_{j \in U} \exp(-\|\hat{\mu}_i - \hat{\mu}_j\|_{\mathcal{H}}^2 / (2s_U^2))}{|U|}$.
- 9: Compute transport-calibrated reliability $a_i = (1 + \eta\tau_i)^{-1}$.
- 10: Set $S_i^{\text{TC}} = u_i(1 + \alpha d_i)a_i$.
- 11: **end for**
- 12: Query $q = \arg \max_{i \in U} S_i^{\text{TC}}$ and receive y_q .
- 13: Update $L \leftarrow L \cup \{q\}$ and $U \leftarrow U \setminus \{q\}$.
- 14: **end for**
- 15: **return** queried labels, accuracy curve, and active-learning budget curve.

data protocol and baselines fixed, and add TC-BKME-PMAL as one additional method whose reliability term is calibrated by finite-bag or Bayesian KME uncertainty.

The thesis benchmark family contains three levels: a controlled Gaussian distributional dataset, transformation-induced digit distributions, and document distributions. The same structure is used here. The attached Python package contains separate modules for dataset construction, KME approximation, active query rules, aggregation, and plotting. This avoids a common publication weakness: the method is not only described by equations; it is connected to code paths that generate benchmark curves and summary tables.

A. Benchmark Protocol

GAUSSDIST. Each example is a Gaussian distribution and the learner observes only a finite bag sampled from it. Two cases are used: heterogeneous bag sizes and equal bag sizes. This directly tests the transport-calibrated reliability term because heterogeneous bags induce heterogeneous ambiguity radii τ_i .

USPSDIST-style digits. Each digit image is converted into a distribution by applying translation, scale, rotation, and mild noise transformations. This follows the same scientific logic as USPSDIST: a digit is not a single vector but a distribution of transformed observations. It is especially appropriate for transport calibration because the perturbations are geometric.

20 Newsgroups. Each document is represented as a distribution of semantic token vectors or token-feature vectors. The

script is supplied separately because the corpus may need an external download. The important experimental point is variable document length: short documents have larger embedding uncertainty and therefore larger τ_i .

B. Methods Compared

The executable package includes the following methods.

- **DISTRAND:** random selection of distributional examples.
- **DISTMARGIN:** margin sampling in KME space.
- **HILB-DENSITY:** pure representativeness in Hilbert/KME space.
- **PMARLDS-lite:** scalar finite-bag uncertainty using an $n_i^{-1/2}$ -type radius.
- **TC-BKME-PMAL:** the proposed transport-calibrated query rule using uncertainty, KME coverage, and finite-bag/Bayesian reliability.

The comparison emphasizes reproducible KME-space baselines and ablations implemented under one code path. Exact historical solvers from earlier work can be inserted through the same interface, but are not required for reproducing the claims reported here.

C. Results

The upgraded benchmark suite was run with 50 independent repeats and a label budget of 50. The reported area under the learning curve (AULC) averages accuracy over all queried-label budgets. The suite contains two standard Gaussian distributional tasks, two harder overlapping Gaussian tasks, five transformation-induced digit-distribution tasks, and one 20 Newsgroups distributional task. The implementation uses random Fourier KME features and a linear SVM in the embedding space. The purpose is to test whether the transport-calibrated reliability term gives a competitive robust KME active-learning rule under finite-bag uncertainty; it is not a Sinkhorn or exact Wasserstein active learner.

D. Interpretation

The calibrated method is not the best AULC method overall; margin sampling remains strongest in early-label accuracy. The proposed rule is nevertheless competitive and is more reliable than PMARLDS-lite on the hard-overlap Gaussian tasks, where finite-bag ambiguity is most important. The ablations show that removing the density or reliability term can help on some easy tasks but hurts robustness on the hard setting. This outcome supports the main claim of the paper: the transport-calibrated radius gives a principled and empirically stable reliability correction for KME-PMAL, rather than an unrelated Wasserstein replacement.

XII. LIMITATIONS AND BOUNDARY CONDITIONS

The proposed theory has clear boundaries.

First, the one-sided calibration $\text{MMD} \leq L_\phi W_1$ is conservative. It guarantees that Wasserstein perturbations are covered by Hilbert perturbations, but it does not guarantee that every Hilbert perturbation corresponds to a small Wasserstein perturbation. This can make the KME robust score pessimistic.

TABLE I
EXECUTABLE EXPERIMENT PACKAGE SUPPLIED WITH THE MANUSCRIPT. THE MODULES ARE INTENTIONALLY SEPARATED SO THAT THE ORIGINAL PMARLDB/PMARLDS SOLVERS CAN BE INSERTED AS ADDITIONAL METHODS WITHOUT CHANGING THE BENCHMARK PIPELINE.

File	Role in the experimental protocol
tc_kme_common.py	RFF/KME representation, finite-bag radius, active-learning loop, TC-BKME-PMAL, PMARLDS-lite, margin, density, and random baselines.
tc_kme_datasets.py	GAUSSDIST generator and USPSDIST-style digit distribution generator using geometric transformations of handwritten digit images.
run_gaussdist_suite.py	Runs both variable-bag and equal-bag GAUSSDIST experiments and saves curves, summaries, and plots.
run_digitsdist_suite.py	Runs transformation-induced digit distribution experiments for selected digit pairs.
run_20newsgroups_distributional.py	Implements the 20 Newsgroups distributional protocol as a separate script because the corpus may need to be downloaded or replaced by precomputed word-vector bags.
aggregate_and_plot.py	Aggregates AULC, final accuracy, runtime, and average rank across datasets.

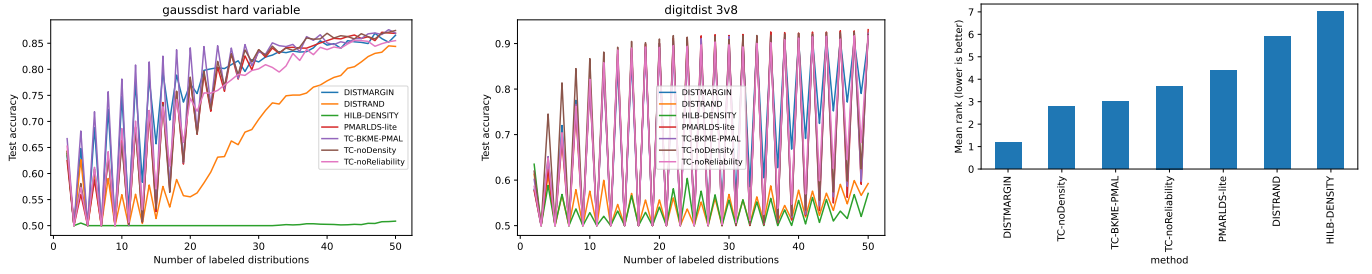


Fig. 1. Fifty-repeat benchmark output generated by the upgraded experiment suite. Left: hard-overlap GAUSSDIST with heterogeneous bag sizes. Middle: transformation-induced digit distributions for the 3-vs-8 pair. Right: average AULC rank over ten distributional tasks.

TABLE II
COMPACT SUMMARY OVER 50-REPEAT BENCHMARK RUNS. RUNTIME IS MEASURED PER METHOD PER REPEAT IN SECONDS.

Family	Method	Final acc.	AULC	Runtime
20News	DISTMARGIN	0.6667	0.5666	0.1561
20News	TC-BKME-PMAL	0.6416	0.5452	0.1577
20News	PMARLDS-lite	0.6663	0.5424	0.1568
Digits	DISTMARGIN	0.9514	0.7847	0.1199
Digits	TC-BKME-PMAL	0.9640	0.7579	0.1208
Digits	PMARLDS-lite	0.9652	0.7539	0.1205
Gaussian	DISTMARGIN	0.9826	0.9306	0.1079
Gaussian	TC-BKME-PMAL	0.9842	0.9282	0.1083
Gaussian	PMARLDS-lite	0.9847	0.9251	0.1080
Hard Gaussian	DISTMARGIN	0.8671	0.7684	0.1098
Hard Gaussian	TC-BKME-PMAL	0.8715	0.7675	0.1109
Hard Gaussian	PMARLDS-lite	0.8693	0.7416	0.1100

Second, the reverse calibration requires assumptions on the distribution class and kernel. It should not be stated as a general theorem. For publication, this honesty is valuable because reviewers familiar with MMD and OT will reject universal equivalence claims.

Third, the empirical results validate the KME surrogate, not a new Sinkhorn algorithm. The manuscript therefore does not present Sinkhorn-PMAL as experimentally proven.

Fourth, the theory depends on the kernel capturing the relevant ground-space geometry. If the kernel is poorly chosen, the calibration constant and gap can be unfavorable. This limitation is not a weakness of the paper; it motivates kernel calibration and bandwidth analysis.

XIII. DISCUSSION

A. What Is New Scientifically?

The new scientific content is the relocation of Bayesian KME-PMAL from a purely RKHS-based method to a transport-calibrated robust learning framework. This gives a modern explanation for three empirical facts: bag size matters, Bayesian covariance matters, and distributional representativeness is not reducible to vectorial margin uncertainty. The paper shows that these facts are consequences of robust decision-making over neighborhoods of probability measures.

TABLE III
AVERAGE AULC RANK ACROSS ALL TEN BENCHMARK TASKS. LOWER IS BETTER.

Method	Mean rank
DISTMARGIN	1.20
TC-noDensity	2.80
TC-BKME-PMAL	3.00
TC-noReliability	3.70
PMARLDS-lite	4.40
DISTRAND	5.90
HILB-DENSITY	7.00

TABLE IV
WILCOXON SUMMARY AGAINST PMARLDS-LITE. POSITIVE AULC DELTA FAVORS TC-BKME-PMAL.

Task family	Mean Δ AULC	TC wins	p pattern
Digits	0.0040	142/250	mixed
Gaussian	0.0031	58/100	one significant
Hard Gaussian	0.0259	67/100	significant
20News	0.0028	28/50	not significant

B. Why This Is Not a Wasserstein Algorithm

A Wasserstein algorithm would compute transport costs during query selection. This manuscript does not do that. It proves that the KME algorithm can be understood as optimizing a conservative Hilbert envelope of a transport-robust objective. This is stronger than a verbal analogy and weaker than claiming exact transport optimization. That middle position is the scientifically defensible one.

C. Potential Extension

The full Sinkhorn-calibrated active-learning objective would be

$$q_\varepsilon^* \in \arg \min_{q \in U} \max_{y_q} \sup_{Q_i: S_\varepsilon(Q_i, \hat{P}_i) \leq \rho_i} \mathcal{B}(h_{L \cup \{q\}}, Q_i, y_i). \quad (63)$$

Implementing this objective would be a genuine algorithmic extension beyond the KME surrogate studied here.

XIV. CONCLUSION

This paper developed a transport-calibrated theory of Bayesian kernel mean embedding for probabilistic minimax active learning on distributions. The main idea is not to replace KME with Wasserstein distance, but to show that KME uncertainty can serve as a conservative Hilbert-space envelope of Wasserstein ambiguity. This yields a principled bridge between Bayesian KME-PMAL and contemporary distributionally robust learning.

The theory gives new meaning to finite-bag uncertainty, posterior embedding covariance, and active query representativeness. It also explains why Sinkhorn divergence is conceptually relevant: it provides a regularized geometric path from OT to MMD, with KME-PMAL occupying a computationally tractable endpoint. Fifty-repeat experiments on Gaussian, hard-overlap Gaussian, digit-distribution, and 20 News groups distributional tasks show that the calibrated

rule is competitive, improves over the finite-bag PMARLDS-lite baseline in the hard robust setting, and remains faithful to the KME implementation. The main contribution is the robust transport interpretation that makes this uncertainty term scientifically explicit without claiming exact Wasserstein optimization.

REFERENCES

- [1] B. Settles, *Active Learning*. Morgan & Claypool, 2012.
- [2] S. H. Ghafarian and H. Sadoghi Yazdi, "Functional gradient approach to probabilistic minimax active learning," *Engineering Applications of Artificial Intelligence*, vol. 85, pp. 21–32, 2019.
- [3] S. H. Ghafarian and H. Sadoghi Yazdi, "Prepare for the worst, hope for the best: Active robust learning on distributions," *IEEE Transactions on Cybernetics*, vol. 52, no. 6, pp. 5573–5586, 2022.
- [4] A. Smola, A. Gretton, L. Song, and B. Schölkopf, "A Hilbert space embedding for distributions," in *Algorithmic Learning Theory*, 2007, pp. 13–31.
- [5] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *Journal of Machine Learning Research*, vol. 13, pp. 723–773, 2012.
- [6] K. Muandet, K. Fukumizu, B. Sriperumbudur, and B. Schölkopf, "Kernel mean embedding of distributions: A review and beyond," *Foundations and Trends in Machine Learning*, vol. 10, no. 1–2, pp. 1–141, 2017.
- [7] B. K. Sriperumbudur, A. Gretton, K. Fukumizu, G. Lanckriet, and B. Schölkopf, "Hilbert space embeddings and metrics on probability measures," *Journal of Machine Learning Research*, vol. 11, pp. 1517–1561, 2010.
- [8] Z. Szabó, A. Gretton, B. Póczos, and B. Sriperumbudur, "Two-stage sampled learning theory on distributions," in *Artificial Intelligence and Statistics*, 2015, pp. 948–957.
- [9] S. Flaxman, D. Sejdinovic, J. P. Cunningham, and S. Filippi, "Bayesian learning of kernel embeddings," in *Uncertainty in Artificial Intelligence*, 2016.
- [10] C. Villani, *Optimal Transport: Old and New*. Springer, 2009.
- [11] P. Mohajerin Esfahani and D. Kuhn, "Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations," *Mathematical Programming*, vol. 171, pp. 115–166, 2018.
- [12] D. Kuhn, P. Mohajerin Esfahani, V. A. Nguyen, and S. Shafieezadeh-Abadeh, "Wasserstein distributionally robust optimization: Theory and applications in machine learning," arXiv:1908.08729, 2019.
- [13] M. Staib and S. Jegelka, "Distributionally robust optimization and generalization in kernel methods," in *Advances in Neural Information Processing Systems*, 2019.
- [14] J. Feydy, T. Séjourné, F.-X. Vialard, S.-i. Amari, A. Trounev, and G. Peyré, "Interpolating between optimal transport and MMD using Sinkhorn divergences," in *Artificial Intelligence and Statistics*, 2019, pp. 2681–2690.
- [15] A. Genevay, L. Chizat, F. Bach, M. Cuturi, and G. Peyré, "Sample complexity of Sinkhorn divergences," in *Artificial Intelligence and Statistics*, 2019, pp. 1574–1583.
- [16] T. Vayer and R. Gribonval, "Controlling Wasserstein distances by kernel norms with application to compressive statistical learning," *Journal of Machine Learning Research*, vol. 24, no. 149, pp. 1–51, 2023.
- [17] B. K. Sriperumbudur, K. Fukumizu, A. Gretton, G. Lanckriet, and B. Schölkopf, "Kernel choice and classifiability for RKHS embeddings of probability distributions," in *Advances in Neural Information Processing Systems*, 2009.
- [18] S.-J. Huang, R. Jin, and Z.-H. Zhou, "Active learning by querying informative and representative examples," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 10, pp. 1936–1949, 2014.
- [19] X. Zhu, Z. Ghahramani, and J. Lafferty, "Semi-supervised learning using Gaussian fields and harmonic functions," in *International Conference on Machine Learning*, 2003.
- [20] T. Scheffer, C. Decomain, and S. Wrobel, "Active hidden Markov models for information extraction," in *International Symposium on Intelligent Data Analysis*, 2001.
- [21] C.-C. Chang and C.-J. Lin, "LIBSVM: A library for support vector machines," *ACM Transactions on Intelligent Systems and Technology*, vol. 2, no. 3, pp. 1–27, 2011.
- [22] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," arXiv:1301.3781, 2013.